

GREAT version 4.0.4 current (08/19/2019 to now)

▼

Warning: Your set hits a large fraction of the genes in the genome, which often does not work well with the GREAT Significant by Both view due to a saturation of the gene-based hypergeor
See our [tips for handling large datasets](#) or try the [Significant By Region-based Binomial](#) view.

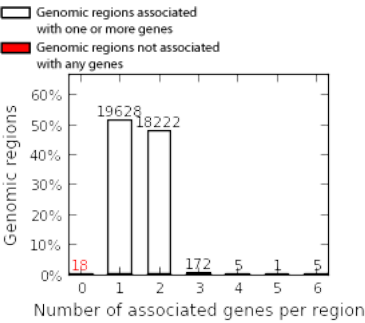
Job Description

Region-Gene Association Graphs

What do these graphs illustrate?

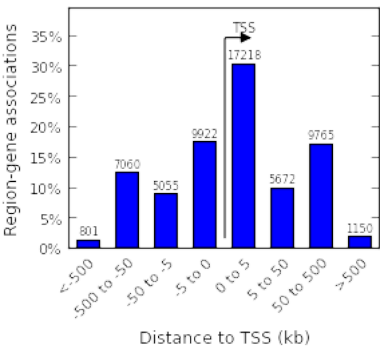
Number of associated genes per region

Download as PDF.



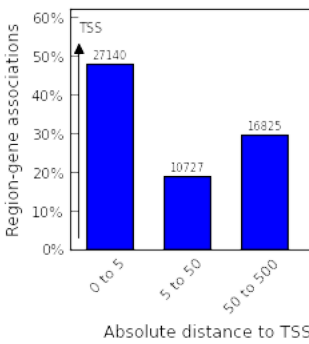
Binned by orientation and distance to TSS

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Binned by absolute distance to TSS

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Global Controls

Global Export ▼

Which data is exported by each option?

Ensembl Genes (no terms)

GO Biological Process (17 terms)

Table controls: Export ▼ Shown top rows in this table: 20 Set Term annotation count: Min: 1 Max: Inf Set Visualize this table: [select one]

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hi
nuclear-transcribed mRNA catabolic process, nonsense-mediated decay	57	3.9728e-37	9.1716e-35	2.0004	414	1.09%	536	1.9906e-6	1.3282	103
intrinsic apoptotic signaling pathway in response to DNA damage	64	1.2562e-35	2.5829e-33	2.1016	351	0.92%	924	8.1227e-4	1.3232	60
snRNA metabolic process	74	7.3696e-33	1.3105e-30	2.0917	326	0.86%	1,476	2.0480e-2	1.2208	69
IRE1-mediated unfolded protein response	77	2.9079e-32	4.9694e-30	2.3195	254	0.67%	1,116	4.5442e-3	1.3118	50
snRNA transcription from RNA polymerase II promoter	82	3.8236e-30	6.1359e-28	2.0789	302	0.79%	1,328	1.1541e-2	1.2608	58
regulation of receptor biosynthetic process	89	1.5503e-28	2.2922e-26	3.1518	131	0.34%	1,500	2.2562e-2	1.4456	19
alpha-linolenic acid metabolic process	101	3.4718e-27	4.5233e-25	4.0813	89	0.23%	1,625	3.4465e-2	1.5216	13
ER-nucleus signaling pathway	157	5.9987e-23	5.0278e-21	2.1840	201	0.53%	1,149	5.2398e-3	1.3912	32
histone monoubiquitination	222	8.8260e-20	5.2316e-18	2.1512	177	0.47%	1,651	3.6356e-2	1.3586	25
intrinsic apoptotic signaling pathway in response to DNA damage by p53 class mediator	254	5.6765e-19	2.9408e-17	2.3468	140	0.37%	1,565	2.7860e-2	1.3642	26
positive regulation of triglyceride metabolic process	349	5.5306e-16	2.0853e-14	2.6336	93	0.24%	1,594	3.0846e-2	1.4416	18
regulation of cholesterol homeostasis	438	1.1702e-13	3.5157e-12	2.5264	84	0.22%	1,750	4.8714e-2	1.5216	12
hematopoietic stem cell differentiation	452	1.8288e-13	5.3243e-12	2.5205	83	0.22%	1,750	4.8714e-2	1.5216	12
'de novo' posttranslational protein folding	713	1.0950e-9	2.0209e-8	2.1567	78	0.20%	1,167	5.8709e-3	1.5216	18
chaperone mediated protein folding requiring cofactor	877	2.0847e-8	3.1280e-7	2.5394	47	0.12%	1,750	4.8714e-2	1.5216	12
proteasome assembly	1,087	4.2890e-7	5.1922e-6	2.1963	51	0.13%	1,625	3.4465e-2	1.5216	13
positive regulation of transcription elongation from RNA polymerase II promoter	1,229	1.8502e-6	1.9811e-5	2.2471	43	0.11%	1,435	1.6844e-2	1.5216	15

13,159 ontology terms (100%) were tested using an annotation count range of [1, Inf].

GO Cellular Component (2 terms)

Table controls:

Export

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Set

Term annotation count: Min:

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 Max:

Inf

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Visualize this table:

[select one]

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits
COPI-coated vesicle	130	1.9323e-10	2.5684e-9	2.0185	100	0.26%	145	7.5852e-4	1.5216	23
proteasome regulatory particle, base subcomplex	263	8.7946e-6	5.7783e-5	2.3798	33	0.09%	242	4.6259e-2	1.5216	12

The test set of 38,051 genomic regions picked 12,340 (66%) of all 18,777 genes.
GO Cellular Component has 1,728 terms covering 17,911 (95%) of all 18,777 genes, and 382,522 term - gene associations.
1,728 ontology terms (100%) were tested using an annotation count range of [1, Inf].

GO Molecular Function (no terms)

Human Phenotype (3 terms)

Table controls:

Export

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Term annotation count: Min:

1

 Max:

Inf

Set

Visualize this table:

[select one]

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits
Abnormal glycosylation	19	7.1656e-16	2.5166e-13	2.6219	93	0.24%	578	1.3980e-2	1.4525	21
Abnormal protein N-linked glycosylation	24	4.0585e-15	1.1284e-12	2.5746	91	0.24%	605	1.9484e-2	1.4492	20
Abnormal isoelectric focusing of serum transferrin	38	1.3779e-13	2.4196e-11	2.4720	87	0.23%	641	2.6774e-2	1.4456	19

The test set of 38,051 genomic regions picked 12,340 (66%) of all 18,777 genes.
Human Phenotype has 6,673 terms covering 3,390 (18%) of all 18,777 genes, and 255,152 term - gene associations.
6,673 ontology terms (100%) were tested using an annotation count range of [1, Inf].

Mouse Phenotype Single KO (3 terms)

Table controls:

Export

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 Max:

Inf

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Visualize this table:

[select one]

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits
delayed embryo turning	114	4.5714e-16	3.6720e-14	2.1422	142	0.37%	921	1.2040e-2	1.4525	21
thin hypoglossal nerve	123	3.7798e-15	2.8140e-13	2.4263	102	0.27%	975	1.7251e-2	1.5216	15
increased squamous cell carcinoma incidence	127	6.0878e-15	4.3894e-13	2.0412	147	0.39%	498	3.3152e-4	1.5216	26

The test set of 38,051 genomic regions picked 12,340 (66%) of all 18,777 genes.
Mouse Phenotype Single KO has 9,157 terms covering 9,525 (51%) of all 18,777 genes, and 563,371 term - gene associations.
9,157 ontology terms (100%) were tested using an annotation count range of [1, Inf].

Mouse Phenotype (4 terms)

Table controls:

Export

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Term annotation count: Min:

1

 Max:

Inf

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Visualize this table:

[select one]

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits
abnormal circulating phospholipid level	138	2.5185e-16	1.7449e-14	2.2152	134	0.35%	1,595	3.1321e-2	1.3893	21
thin hypoglossal nerve	162	3.7798e-15	2.2308e-13	2.4263	102	0.27%	1,324	1.3264e-2	1.5216	15
abnormal frontal lobe morphology	297	8.6792e-11	2.7940e-9	2.0258	103	0.27%	1,324	1.3264e-2	1.5216	15
increased keratinocyte apoptosis	619	3.6842e-7	5.6905e-6	2.0317	62	0.16%	1,523	2.6719e-2	1.5216	13

The test set of 38,051 genomic regions picked 12,340 (66%) of all 18,777 genes.
Mouse Phenotype has 9,561 terms covering 9,709 (52%) of all 18,777 genes, and 718,378 term - gene associations.
9,561 ontology terms (100%) were tested using an annotation count range of [1, Inf].

